Programming for Data Analysis
Assignment Report
Using Python and R

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Contents

1 Introduction

Data analysis is the process of collecting, organizing, exploring, and interpreting data to extract meaningful insights. Python and R are two of the most widely used programming languages for data analysis because they provide powerful tools for handling datasets, performing statistical analysis, and creating visualizations.

In this report, both Python and R were used to perform three basic data analysis tasks:

- Loading and exploring a CSV dataset.
- Creating new columns using existing data.
- Sorting the dataset based on a selected column.

These assignments provide a practical understanding of basic data manipulation techniques that are commonly used in real-world data analysis projects.

2 Dataset

The **Students Performance in Exams** dataset was used in this report.

Dataset Name: Students Performance in Exams

Dataset Source: https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv

The dataset contains eight columns:

- gender
- race/ethnicity
- parental level of education
- lunch
- test preparation course
- math score
- reading score
- writing score

The dataset consists of 1000 observations and 8 variables.

3 Assignment 1: Load and Explore Data (Python)

3.1 Objective

The objective of this assignment is to learn how to load a CSV dataset from the Internet and perform basic data exploration using Python and the Pandas library.

3.2 Python Code

```
import pandas as pd

# Load CSV dataset from the internet
url =
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df = pd.read_csv(url)

# Display first 5 rows
print("First 5 Rows:")
print(df.head())

# Display column names
print("\nColumn Names:")
print(df.columns.tolist())

# Display number of rows and columns
print("\nDataset Shape:")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

# Display summary statistics
print("\nSummary Statistics:")
print(df.describe())
```

3.3 Output

```
First 5 Rows:
   gender race/ethnicity parental level of education  ... math score reading score writing sco
0  female        group B           bachelor's degree  ...         72            72
1  female        group C                some college  ...         69            90
2  female        group B             master's degree  ...         90            95
3    male        group A          associate's degree  ...         47            57
4    male        group C                some college  ...         76            78

Column Names:
['gender', 'race/ethnicity', 'parental level of education', 'lunch',
 'test preparation course', 'math score', 'reading score', 'writing score']

Dataset Shape:
```

```
Rows: 1000
Columns: 8

Summary Statistics:
       math score  reading score  writing score
count  1000.00000    1000.000000    1000.000000
mean     66.08900      69.169000      68.054000
std      15.16308      14.600192      15.195657
min       0.00000      17.000000      10.000000
25%      57.00000      59.000000      57.750000
50%      66.00000      70.000000      69.000000
75%      77.00000      79.000000      79.000000
max     100.00000     100.000000     100.000000
```

3.4 Explanation

In this assignment, the dataset was loaded from an online CSV file using `read_csv()`. The `head()` function showed the first 5 rows, `columns` gave the column names, and `shape` returned the row and column counts (1000 x 8). Finally, `describe()` generated summary statistics – count, mean, standard deviation, min, max, and quartiles – for the three numeric score columns. This gives a quick first look at the dataset before deeper analysis.

4 Assignment 1: Load and Explore Data (R)

4.1 Objective

The objective of this assignment is to learn how to load a CSV dataset from the Internet and perform basic data exploration using the R programming language.

4.2 R Code

```r
# Load CSV dataset from the internet
url <-
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df <- read.csv(url)

# Display first 5 rows
cat("First 5 Rows:\n")
head(df, 5)

# Display column names
cat("\nColumn Names:\n")
colnames(df)

# Display number of rows and columns
cat("\nDataset Shape:\n")
cat("Rows:", nrow(df), "\n")
cat("Columns:", ncol(df), "\n")

# Display summary statistics
```

```r
cat("\nSummary Statistics:\n")
summary(df)
```

4.3 Output

```
First 5 Rows:
  gender race.ethnicity parental.level.of.education        lunch
1 female        group B           bachelor's degree     standard
2 female        group C                some college     standard
3 female        group B             master's degree     standard
4   male        group A          associate's degree free/reduced
5   male        group C                some college     standard
  test.preparation.course math.score reading.score writing.score
1                    none         72            72            74
2               completed         69            90            88
3                    none         90            95            93
4                    none         47            57            44
5                    none         76            78            75

Column Names:
[1] "gender"                       "race.ethnicity"
[3] "parental.level.of.education"  "lunch"
[5] "test.preparation.course"      "math.score"
[7] "reading.score"                "writing.score"

Dataset Shape:
Rows: 1000
Columns: 8

Summary Statistics:
   math.score     reading.score    writing.score
 Min.   :  0.00   Min.   : 17.00   Min.   : 10.00
 1st Qu.: 57.00   1st Qu.: 59.00   1st Qu.: 57.75
 Median : 66.00   Median : 70.00   Median : 69.00
 Mean   : 66.09   Mean   : 69.17   Mean   : 68.05
 3rd Qu.: 77.00   3rd Qu.: 79.00   3rd Qu.: 79.00
 Max.   :100.00   Max.   :100.00   Max.   :100.00
```

4.4 Explanation

In this assignment, the same dataset was loaded in R using `read.csv()`. The `head()`, `colnames()`, `nrow()`, and `ncol()` functions reproduced the same exploration steps as Python, and `summary()` generated the descriptive statistics. The only real difference is that R automatically converts spaces and slashes in column names to dots (e.g., `math score` becomes `math.score`), so the code refers to columns slightly differently than in Python. All numeric results matched exactly.

5 Assignment 2: Creating New Columns (Python)

5.1 Objective

The objective of this assignment is to learn how to create new columns in a dataset using existing data and apply conditional statements in Python using the Pandas library.

5.2 Python Code

```
import pandas as pd

url =
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df = pd.read_csv(url)

# Create Average Score column
df["Average Score"] = (
    df["math score"] + df["reading score"] + df["writing score"]
) / 3

# Apply condition
df["Result"] = df["Average Score"].apply(
    lambda x: "Pass" if x >= 60 else "Fail"
)

print(df[["math score", "reading score", "writing score",
          "Average Score", "Result"]].head())
print()
print(df["Result"].value_counts())
```

5.3 Output

```
   math score  reading score  writing score  Average Score Result
0          72             72             74      72.666667   Pass
1          69             90             88      82.333333   Pass
2          90             95             93      92.666667   Pass
3          47             57             44      49.333333   Fail
4          76             78             75      76.333333   Pass

Result
Pass    715
Fail    285
Name: count, dtype: int64
```

5.4 Explanation

In this assignment, a new column, Average Score, was created by averaging the three score columns. A second column, Result, was then generated using `apply()` with a lambda function, labeling students as Pass or Fail based on a 60-point cutoff. Out of 1000 students, 715 passed and 285 failed. This demonstrates basic feature engineering – turning raw columns into a new, more useful variable.

6 Assignment 2: Creating New Columns (R)

6.1 Objective

The objective of this assignment is to learn how to create new columns using existing data and apply conditional statements in R.

6.2 R Code

```r
url <-
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df <- read.csv(url)

# Create Average Score column
df$Average.Score <- (
  df$math.score + df$reading.score + df$writing.score
) / 3

# Create Result column
df$Result <- ifelse(
  df$Average.Score >= 60,
  "Pass",
  "Fail"
)

# Display first 6 rows
head(df[, c("math.score", "reading.score", "writing.score",
            "Average.Score", "Result")])
table(df$Result)
```

6.3 Output

```
  math.score reading.score writing.score Average.Score Result
1         72            72            74      72.66667   Pass
2         69            90            88      82.33333   Pass
3         90            95            93      92.66667   Pass
4         47            57            44      49.33333   Fail
5         76            78            75      76.33333   Pass
6         71            83            78      77.33333   Pass

Fail Pass
 285  715
```

6.4 Explanation

In this assignment, the same Average.Score column was created using R's vectorized arithmetic, and `ifelse()` was used to generate the Result column with the same 60-point Pass/Fail rule. The output matched Python exactly (285 Fail, 715 Pass), confirming both implementations are correct. `ifelse()` is R's element-wise equivalent of Python's `apply()` + lambda pattern.

7 Assignment 3: Sorting Data (Python)

7.1 Objective

The objective of this assignment is to learn how to sort a dataset based on a specific column using Python and the Pandas library.

7.2 Python Code

```python
import pandas as pd

url =
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df = pd.read_csv(url)

df["Average Score"] = (
    df["math score"] + df["reading score"] + df["writing score"]
) / 3

# Sort dataset by math score
sorted_df = df.sort_values(
    by="math score",
    ascending=True
)

print(sorted_df[["math score", "reading score", "writing score"]].head(10))
```

7.3 Output

```
     math score  reading score  writing score
59            0             17             10
980           8             24             23
17           18             32             28
787          19             38             32
145          22             39             33
842          23             44             36
338          24             38             27
466          26             31             38
363          27             34             32
91           27             34             36
```

7.4 Explanation

In this assignment, the dataset was sorted in ascending order by math score using `sort_values()`. The original row index (e.g., 59, 980, 17. . .) stays attached to each row even after sorting, so the lowest scorers can be traced back to their original position. This makes it easy to spot outliers – such as the student with a math score of 0 – for further investigation.

8 Assignment 3: Sorting Data (R)

8.1 Objective

The objective of this assignment is to learn how to sort a dataset based on a selected column using the R programming language.

8.2 R Code

```r
url <-
↪  "https://raw.githubusercontent.com/rashida048/Datasets/master/StudentsPerformance.csv"
df <- read.csv(url)

df$Average.Score <- (
  df$math.score + df$reading.score + df$writing.score
) / 3

# Sort the dataset in ascending order
sorted_data <- df[order(df$math.score), ]

# Display first 10 rows
head(sorted_data[, c("math.score", "reading.score", "writing.score")], 10)
```

8.3 Output

```
    math.score reading.score writing.score
60           0            17            10
981          8            24            23
18          18            32            28
788         19            38            32
146         22            39            33
843         23            44            36
339         24            38            27
467         26            31            38
92          27            34            36
364         27            34            32
```

8.4 Explanation

In this assignment, the same ascending sort was performed in R using `order()`, which returns the row order rather than sorting in place. The results matched Python's output exactly, aside from a one-row index offset – R starts row numbering at 1, while Python starts at 0. This confirms both languages sorted the data identically.

9 Summary of Results

The following table summarizes the tasks completed in this report.

Task	Status
Dataset Loaded Successfully	Completed
Data Exploration	Completed
Summary Statistics Generated	Completed
New Column (Average Score) Created	Completed
Conditional Column (Result) Created	Completed
Dataset Sorted Successfully	Completed
Python Implementation	Completed
R Implementation	Completed

The assignments were successfully completed using both Python and R. The Students Performance dataset was loaded, explored, modified, and sorted according to the assignment requirements. The outputs generated by both programming languages were consistent, demonstrating that the same data analysis tasks can be performed effectively in either language.

10 Conclusion

This report provided practical experience in performing basic data analysis using both Python and R. The tasks included importing a dataset from an online source, exploring its structure, generating summary statistics, creating new columns, applying conditional statements, and sorting the dataset based on a calculated value.

Python, through the Pandas library, offers powerful and efficient tools for data manipulation, while R provides simple and effective functions for statistical analysis and data processing. Although the syntax of the two programming languages is different, both can accomplish the same analytical tasks successfully.

Overall, this report improved my understanding of fundamental data analysis concepts and strengthened my programming skills in Python and R. These techniques will serve as a foundation for more advanced topics such as data visualization, machine learning, and predictive analytics.

11 References

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